

1. (a) Let $u = \arcsin x \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$; $dv = x dx \Rightarrow v = \frac{x^2}{2}$

$$\int x \arcsin x dx = \frac{x^2}{2} \arcsin x - \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} dx$$

Let $x = \sin u \Rightarrow dx = \cos u du$

$$\int x \arcsin x dx = \frac{x^2}{2} \arcsin x - \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} dx = \frac{x^2}{2} \arcsin x - \frac{1}{2} \int \sin^2 u du$$

$$= \frac{x^2}{2} \arcsin x - \frac{1}{2} \left[\frac{1}{2} u - \frac{1}{4} \sin 2u \right] + C = \frac{x^2}{2} \arcsin x - \frac{1}{4} \arcsin x + \frac{1}{8} (2x \sqrt{1-x^2}) + C$$

$$= \arcsin \left(\frac{x^2}{2} - \frac{1}{4} \right) + \frac{1}{4} x \sqrt{1-x^2} + C$$
- (b)
$$\int \frac{dx}{(x+1)(x+2)(x+3)} = \int \left(-\frac{1}{2} \frac{1}{x+1} + \frac{2}{x+2} - \frac{3}{2} \frac{1}{x+3} \right) dx$$

$$= -\frac{1}{2} \ln|x+1| + 2 \ln|x+2| - \frac{3}{2} \ln|x+3| + C$$
- (c) Let $x+2 = 3 \tan u \Rightarrow dx = 3 \sec^2 u du$

$$\int \frac{x+2}{\sqrt{x^2+4x+13}} dx = \int \frac{x+2}{\sqrt{(x+2)^2+9}} dx = \int \frac{3 \tan u}{3 \sec u} \cdot 3 \sec^2 u du$$

$$= 3 \int \sec u \tan u du = 3 \sec u + C = \sqrt{x^2+4x+13} + C$$
- (d)
$$\int \frac{x^2+1}{x(x^2-1)} = \int \left(\frac{-1}{x} + \frac{1}{x-1} + \frac{1}{x+1} \right) dx$$

$$= -\ln|x| + \ln|x-1| + \ln|x+1| + C = \ln \left| \frac{x^2-1}{x} \right| + C$$
- (e) Let $x = \sec u \Rightarrow dx = \sec u \tan u du$

$$\int \frac{\sqrt{x^2-1}}{x} dx = \int \frac{\tan u}{\sec u} \sec u \tan u du = \int \tan^2 u du = \int (\sec^2 u - 1) du$$

$$= \tan u - u + C = \sqrt{x^2-1} - \arctan \sqrt{x^2-1} + C$$
- (f)
$$\int \sin^3 x \cos^2 x dx = \int \cos^2 (1 - \cos^2 x) \sin x dx = -\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x + C$$
- (g) Let $\frac{x}{4} = \tan u \Rightarrow dx = 4 \sec^2 u du$

$$\int \frac{dx}{(x^2+16)^2} = \frac{1}{64} \int \cos^2 u du = \frac{1}{128} \left[u + \frac{\sin 2u}{2} \right] + C = \frac{1}{128} \arctan\left(\frac{x}{4}\right) + \frac{x}{32(x^2+16)} + C$$
- (h)
$$\int \frac{1}{(x-1)(x^2+1)^2} = \frac{1}{4} \int \left[\frac{1}{x-1} - \frac{x+1}{x^2+1} - \frac{2x+2}{(x^2+1)^2} \right] dx$$

$$= \frac{1}{4} \int \left[\frac{1}{x-1} - \frac{x}{x^2+1} - \frac{1}{x^2+1} - \frac{2x}{(x^2+1)^2} - \frac{2}{(x^2+1)^2} \right] dx$$

$$= \frac{1}{4} \ln|x-1| - \frac{1}{8} \ln(x^2+1) - \frac{1}{4} \arctan x + \frac{1}{4(x^2+1)} - \frac{1}{4} \left(\arctan x + \frac{x}{x^2+1} \right) + C$$

$$= \frac{1}{8} \ln \left(\frac{x^2-2x+1}{x^2+1} \right) - \frac{1}{2} \arctan x - \frac{x-1}{4(x^2+1)} + C$$
- (i)
$$\int \sin^2 x \sin 2x dx = \int \sin^2 x (2x \sin x \cos x) dx = 2 \int \sin^3 x \cos x dx = \frac{1}{2} \sin^4 x + C$$
- (j)
$$\int \frac{x^2-4x+3}{x^3+2x^2+x} dx = \int \frac{x^2-4x+3}{x(x+1)^2} dx = \int \left(\frac{3}{x} - \frac{2}{x+1} - \frac{8}{(x+1)^2} \right) dx$$

$$= 3 \ln|x| - 2 \ln|x+1| + \frac{8}{x+1} + C$$

(k) Let $x - 1 = 2 \sec u \Rightarrow dx = 2 \sec u \tan u du$

$$\int \frac{dx}{\sqrt{x^2 - 2x - 3}} = \int \frac{dx}{\sqrt{(x-1)^2 - 4}} = \int \frac{2 \sec u \tan u}{2 \tan u} du = \int \sec u du$$

$$= \ln |\sec u + \tan u| + C = \ln |x - 1 + \sqrt{(x-1)^2 - 4}| + C$$

$$(l) \int \frac{\cos \theta}{\sin^2 \theta - 2 \sin \theta - 8} d\theta = \frac{1}{6} \int \frac{\cos \theta}{\sin \theta - 4} d\theta - \frac{1}{6} \int \frac{\cos \theta}{\sin \theta + 2} d\theta$$

$$= \frac{1}{6} \ln |\sin \theta - 4| - \frac{1}{6} \ln |\sin \theta + 2| + C = \frac{1}{6} \ln \left| \frac{\sin \theta - 4}{\sin \theta + 2} \right| + C$$

$$(m) \int \frac{1}{t([\ln t]^2 - 4)} dt = \frac{1}{4} \int \frac{1}{t(\ln t - 2)} dt - \frac{1}{4} \int \frac{1}{t(\ln t + 2)} dt$$

$$= \frac{1}{4} \ln |\ln t - 2| - \frac{1}{4} \ln |\ln t + 2| + C = \frac{1}{4} \ln \left| \frac{\ln t - 2}{\ln t + 2} \right| + C$$

$$(n) \int (\sin 3x - \sin x)^2 dx = \int (\sin^2 3x - 2 \sin 3x \sin x + \sin^2 x) dx$$

$$= \int \left(\frac{1}{2} - \frac{1}{2} \cos 6x - \cos 2x + \cos 4x + \frac{1}{2} - \frac{1}{2} \cos 2x \right) dx$$

$$= x - \frac{1}{12} \sin 6x + \frac{1}{4} \sin 4x - \frac{3}{4} \sin 2x + C$$

$$(o) \int \frac{1}{1 + \sin x + \cos x} dx = \int \frac{2}{1 + \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}} \cdot \frac{2}{1+u^2} du$$

$$= \int \frac{2}{2+2u} du = \int \frac{du}{1+u} = \ln |1+u| + C = \ln |1 + \tan(\frac{x}{2})| + C$$

$$(p) \int \frac{\sec^2 \theta}{\tan^3 \theta - \tan^2 \theta} d\theta = \int \frac{\sec^2 \theta}{\tan \theta - 1} d\theta - \int \frac{\sec^2 \theta}{\tan \theta} d\theta - \int \frac{\sec^2 \theta}{\tan^2 \theta} d\theta$$

$$= \ln |\tan \theta - 1| - \ln |\tan \theta| + \cot \theta + C = \ln \left| \frac{\tan \theta - 1}{\tan \theta} \right| + \cot \theta + C$$

$$(q) \text{ Let } u = \arctan x \Rightarrow du = \frac{1}{x^2 + 1} dx ; dv = x^3 dx \Rightarrow v = \frac{x^4}{4}$$

$$\int x^3 \arctan x dx = \frac{x^4}{4} \arctan x - \frac{1}{4} \int \frac{x^4}{x^2 + 1} dx = \frac{x^4}{4} \arctan x - \frac{1}{4} \int \left(x^2 - 1 + \frac{1}{x^2 + 1} \right) dx$$

$$= \frac{x^4}{4} \arctan x - \frac{1}{4} \left(\frac{x^3}{3} - x + \arctan x \right) + C = \frac{1}{4} \arctan x (x^4 - 1) - \frac{1}{4} \left(\frac{x^3}{3} - x \right) + C$$

$$(r) \int \csc^2 2x dx = -\frac{1}{2} \cot 2x + C$$

$$(s) \int \cos 2x \sin 3x dx = \int \frac{1}{2} [\sin(3x - 2x) + \sin(3x + 2x)] dx$$

$$= \frac{1}{2} \int (\sin x + \sin 5x) dx = -\frac{1}{2} \cos x - \frac{1}{10} \cos 5x + C$$

$$(t) \int \tan^3 x \sec^3 x dx = \int (\sec^2 x - 1) \sec^2 x \sec x \tan x dx = \frac{1}{5} \sec^5 x - \frac{1}{3} \sec x + C$$

(u) Let $u^2 = 4x + 1 \Rightarrow 2u du = 4dx$

$$\int \frac{2x^2}{(4x+1)^{5/2}} dx = \int 2 \left(\frac{u^2 - 1}{4} \right)^2 (u^{-5}) \frac{u}{2} du = \frac{1}{16} \int (1 - 2u^{-2} + u^{-4}) du$$

$$= \frac{1}{16} u + \frac{1}{8} u^{-1} - \frac{1}{48} u^{-3} + C = \frac{1}{16} (4x+1)^{1/2} + \frac{1}{8} (4x+1)^{-1/2} - \frac{1}{48} (4x+1)^{-3/2} + C$$

(v) Let $u = e^x \Rightarrow du = e^x dx$

$$\int \frac{1 - e^x}{1 + e^x} dx = \int \frac{1 - u}{u(1 + u)} du = \int \left(\frac{1}{u} - \frac{2}{1 + u} \right) du$$

$$= \ln|u| - 2 \ln|1 + u| + C = x - 2 \ln(1 + e^x) + C$$

(w) Let $u = \tan\left(\frac{x}{2}\right) \Rightarrow dx = \frac{2}{1 + u^2} du$, $\sin x = \frac{2u}{1 + u^2}$

$$\int \frac{\sin x}{1 + \sin^2 x} dx = \int \frac{\frac{2u}{1 + u^2}}{1 + \frac{4u^2}{(1 + u^2)^2}} \cdot \frac{2}{1 + u^2} du = \int \frac{4u}{1 + 6u^2 + u^4} du$$

$$\text{Let } v = u^2 \Rightarrow dv = 2u du$$

$$\int \frac{\sin x}{1 + \sin^2 x} dx = \int \frac{4u}{1 + 6u^2 + u^4} du = \int \frac{2}{1 + 6v + v^2} dv = \int \frac{2}{(v + 3)^2 - 8} dv$$

$$= \frac{\sqrt{2}}{2} \ln \left| \frac{v + 3 - 2\sqrt{2}}{\sqrt{(v + 3)^2 - 8}} \right| + C = \frac{\sqrt{2}}{2} \ln \left| \frac{\tan^2(\frac{x}{2}) + 3 - 2\sqrt{2}}{\sqrt{[\tan^2(\frac{x}{2}) + 3]^2 - 8}} \right| + C$$

(x) Let $u = \cos(\ln x) \Rightarrow du = -\frac{\sin(\ln x)}{x} dx$; $dv = dx \Rightarrow v = x$

$$\int \cos(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx$$

Let $u = \sin(\ln x) \Rightarrow du = \frac{\cos(\ln x)}{x} dx$; $dv = dx \Rightarrow v = x$

$$\int \cos(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx = x \cos(\ln x) + x \sin(\ln x) - \int \cos(\ln x) dx$$

$$\Rightarrow 2 \int \cos(\ln x) dx = x \cos(\ln x) + x \sin(\ln x)$$

$$\Rightarrow \int \cos(\ln x) dx = \frac{1}{2} x [\cos(\ln x) + \sin(\ln x)] + C$$

(y) $\int (e^x + 2x)^2 dx = \int (e^{2x} + 4xe^x + 4x^2) dx = \frac{1}{2} e^{2x} + \frac{4}{3} x^3 + \int 4xe^x dx$

$$= \frac{1}{2} e^{2x} + \frac{4}{3} x^3 + 4xe^x - \int 4e^x dx = \frac{1}{2} e^{2x} + \frac{4}{3} x^3 + 4xe^x - 4e^x + C$$

(z) Let $u = \arcsin 2x \Rightarrow du = \frac{2}{\sqrt{1 - 4x^2}} dx$ $\int \frac{\arcsin 2x}{\sqrt{1 - 4x^2}} dx = \frac{1}{2} \int u du$

$$= \frac{1}{4} u^2 + C = \frac{1}{4} (\arcsin 2x)^2 + C$$

2. Let $x = \tan u \Rightarrow dx = \sec^2 u du$
and $x = 0 \Rightarrow u = 0$; $x = 1 \Rightarrow u = \frac{\pi}{4}$

$$\int_0^1 \left[\frac{1}{(1 + x^2)} \right]^2 \pi dx$$

$$= \pi \int_0^{\frac{\pi}{4}} \cos^2 u du$$

$$= \pi \int_0^{\frac{\pi}{4}} \left(\frac{1}{2} + \frac{1}{2} \cos 2u \right) du$$

$$= \pi \left[\frac{1}{2} u + \frac{1}{4} \sin 2u \right]_0^{\frac{\pi}{4}}$$

$$= \pi \left[\frac{1}{4} + \frac{\pi}{8} \right]$$

$$= \frac{\pi}{4} + \frac{\pi^2}{8}$$

3. * **Correct the statement** $y = \frac{1}{x^2 - 1}$ **to** $y = \frac{1}{x^2 - 1}$

Note that $y = \frac{1}{x^2 - 1} = \frac{1}{2} \left[\frac{1}{x - 1} - \frac{1}{x + 1} \right]$

and thus, $\frac{d^0 y}{dx^0} = \left(\frac{1}{2} \right) (-1)^{0+1} \left[\frac{1}{(x-1)^{0+1}} - \frac{1}{(x+1)^{0+1}} \right]$

Suppose that the statement holds for $n = k - 1$,

i.e. $\frac{d^{k-1} y}{dx^{k-1}} = \left(\frac{1}{2} \right) (-1)^{k-1} (k-1)! \left[\frac{1}{(x-1)^{k-1+1}} - \frac{1}{(x+1)^{k-1+1}} \right] = \left(\frac{1}{2} \right) (-1)^{k-1} (k-1)! \left[\frac{1}{(x-1)^k} - \frac{1}{(x+1)^k} \right]$

then, for $n = k$

$$\begin{aligned} \frac{d^k y}{dx^k} &= \frac{d}{dx} \left(\frac{d^{k-1} y}{dx^{k-1}} \right) = \frac{d}{dx} \left(\left(\frac{1}{2} \right) (-1)^{k-1} (k-1)! \left[\frac{1}{(x-1)^k} - \frac{1}{(x+1)^k} \right] \right) \\ &= \frac{1}{2} (-1)^{k-1} (k-1)! \frac{d}{dx} \left[\frac{1}{(x-1)^k} - \frac{1}{(x+1)^k} \right] \\ &= \frac{1}{2} (-1)^{k-1} (k-1)! \cdot -k \left[\frac{1}{(x-1)^{k+1}} - \frac{1}{(x+1)^{k+1}} \right] \\ &= \frac{1}{2} (-1)^{k-1} (k-1)! \frac{d}{dx} \left[\frac{1}{(x-1)^k} - \frac{1}{(x+1)^k} \right] \\ &= \frac{1}{2} (-1)^k k! \left[\frac{1}{(x-1)^{k+1}} - \frac{1}{(x+1)^{k+1}} \right] \end{aligned}$$

4. (a) Set $u = \sqrt{a^2 - x^2}$

Then $u^2 = a^2 - x^2 \Rightarrow 2u du = -2x dx$

$$\begin{aligned} \int \frac{\sqrt{a^2 - x^2}}{x} dx &= \int \frac{\sqrt{a^2 - x^2}}{x^2} x dx = - \int \frac{u}{a^2 - u^2} du = \int \frac{u^2}{u^2 - a^2} du = \int \left(1 - \frac{a^2}{a^2 - u^2} \right) du \\ &= u + \frac{a}{2} \ln \left| \frac{a - u}{a + u} \right| + C = \sqrt{a^2 - x^2} + \frac{a}{2} \ln \left| \frac{a - \sqrt{a^2 - x^2}}{a + \sqrt{a^2 - x^2}} \right| + C \end{aligned}$$

- (b) Set $u = a \sin u$

Then $dx = a \cos u du$ and $\sqrt{a^2 - x^2} = a \cos u$

$$\begin{aligned} \int \frac{\sqrt{a^2 - x^2}}{x} dx &= \int \frac{a \cos u}{a \sin u} a \cos u du = a \int \frac{\cos^2 u}{\sin u} du = a \int \frac{1 - \sin^2 u}{\sin u} du \\ &= a \int (\csc u - \sin u) du = a \ln |\csc u - \cot u| + a \cos u + C \\ &= \sqrt{a^2 - x^2} + a \ln \left| \frac{a}{x} - \frac{\sqrt{a^2 - x^2}}{x} \right| + C = \sqrt{a^2 - x^2} + a \ln \left| \frac{a - \sqrt{a^2 - x^2}}{x} \right| + C \end{aligned}$$

(c) $\frac{a - \sqrt{a^2 - x^2}}{a + \sqrt{a^2 - x^2}} = \left(\frac{a - \sqrt{a^2 - x^2}}{x} \right)^2$